FEG CHALLENGE 3

Predict the Next Game. Prepare It Before the Click.

Round 1 — Product, Design & Architecture Direction

Scope: PSK web and mobile web

> Use the player’s lobby time to safely prepare the likely next game, then launch the normal provider game unchanged.

Current status: browser mechanism validated locally; full-title staging validation is next.

1. The problem

Game loading starts when player patience is lowest

Today, most game bootstrap work starts **after** the player chooses a title. Network transfer, startup scripts and visual assets then compete on the launch path.

- **MEASURED from FEG-provided PSK data:** players average 2.43 games per session and 16.14 sessions over the analysed period.
- **MEASURED from FEG-provided data:** five providers represent 69.8% of stake, creating a focused integration opportunity.
- A slow or blank transition appears at the moment of highest player intent.
- We cannot solve this by bypassing authorization or rewriting certified game code.

The question

Can the lobby anticipate one likely next game and move safe download work before selection?

2. What the existing HARs tell us

Warm browser state materially changes the network journey

For the same exact final 16-request asset batch:

Measurement	Cold capture	Repeat/warm capture
Full HAR span	**76.358 s**	**16.253 s**
Exact 16-request milestone	**35.568 s**	**6.714 s**
Selected-batch wire bytes	**716,931**	**0**
Selected-batch cache markers	No positive marker	All 16 marked disk cache
Full-capture transfer	**~16.6 MB**	**~12 KB**

The full HAR span was **78.7% shorter**. The selected asset milestone completed **81.1% earlier**, or approximately **5.3×**, in the repeat capture.

Why there are two timings

- **Full HAR span** includes every recorded request, including sparse later fetch/ping traffic.
- **Exact asset milestone** stops when the same exact 16 game-asset requests have completed.
- Both are capture-relative network measurements—not click-to-interactive.
- This is a historical cold/repeat comparison, not a result caused by our prototype.
- Claims such as “6 seconds to 1 second” remain **unvalidated**.

3. Our product direction

Prefetch the user's likely next game—not the whole catalogue

Candidate selection

The lobby chooses **one bounded candidate** from approved signals such as:

- sustained hover, focus or game-details dwell;
- recently played title;
- an existing favourite;
- an existing platform recommendation output.

The selection policy affects speculative requests only. It does **not** change lobby ordering, styling, recommendations or player focus.

Safe preparation

For that candidate, the lobby resolves the exact title build, locale and device tier, then prepares only:

`PRELOADER` → `COMMON` → `SPLASH` → proven critical `PRIMARY`

`SECONDARY` content remains demand-loaded.

Launch

When the player selects the candidate, the normal provider iframe launches unchanged and may reuse the browser's existing HTTP-cache entries.

4. The experience in one line

Authorize → Select → Govern → Warm → Launch → Measure

```
Mandatory authorization succeeds
              ↓
Choose one likely next game
              ↓
Check consent, connection, visibility and byte budget
              ↓
Warm a small exact set of credential-free static assets
              ↓
Player clicks; normal provider iframe launches unchanged
              ↓
Measure exact reuse and the agreed readiness milestone
```

If anything goes wrong

Denial, timeout, unknown authorization, poor connection, Save-Data, wrong variant, budget exhaustion or request failure leads to **no warming and the normal safe path**.

The optimization may be skipped. Mandatory protection never is.

5. Why this approach

Small integration surface; clear safety boundaries

Decision	Why we chose it
Browser HTTP cache	It is the same standard cache the iframe can use
Web/mobile-web only	Matches the PSK delivery surface
One predicted candidate	Limits wasted data and keeps measurement clear
Exact versioned URLs	Browser cache reuse depends on exact identity
Maximum two requests at once	Avoids competing with foreground lobby activity
No certified game changes	Keeps provider behavior and launch authority intact
Normal launch fallback	Prefetch failure cannot break the player journey

Deliberately excluded

No native SDK, service worker, custom cache, provider-code fork, whole-catalogue prefetch or authorization bypass.

6. What is validated—and what is next

Evidence today

- **MEASURED locally:** a completed parent fetch of one unchanged 1,292,928-byte Empire of Gold asset was reused by a later iframe request in Chromium.
- **MEASURED locally:** 2/2 treatment iframe requests transferred zero response-body bytes from the local origin.
- Exact-URL, version-mismatch and `no-store` controls behaved as expected.
- A tested simulated prototype covers authorization order, exact variant selection, byte budgets, cancellation and maximum-two concurrency.

Not yet validated

- Full Empire of Gold prefetch and launch.
- User-game prediction hit rate.
- Staging CORS, cache policy and exact manifest behavior.
- Prototype-caused click-to-ready or input-accepted improvement.
- Any “6 seconds to 1 second” outcome.

Staging update

Access is expected soon but has not yet been validated. Staging is where the design moves from local mechanism evidence to a full-title causal test.

7. How we will prove it on staging

Same launch; only warming changes

CONTROL

- Fresh isolated browser state.
- Same game, browser, locale, device tier and readiness milestone.
- No prefetch before selection.

TREATMENT

- A separate fresh isolated browser state.
- Same conditions as CONTROL.
- Authorization succeeds first.
- The prototype warms the exact approved assets before selection.

We will report

- prediction hit or miss;
- warmed assets and speculative bytes;
- launch transfer and confirmed cache reuse;
- asset milestone and authoritative input-accepted time, if exposed;
- browser, title, provider and number of runs;
- wasted bytes, failures and fallback behavior.

Go/no-go: no player-facing enablement until exact reuse and an agreed launch benefit are repeatable.

8. Team split: A / B / C / D

Four owners, one evidence path

Member	Role	Primary ownership
A	Product & Architecture Lead	Scope, candidate policy, decisions, integration and pitch
B	Browser Systems Lead	Governor, exact asset preparation, normal iframe handoff and fallback
C	Evidence & Experiment Lead	HAR analysis, staging CONTROL/TREATMENT, cache attribution and privacy-safe results
D	Experience, Compliance & QA Lead	Player states, accessibility, failure rehearsal, claim review and demo quality

Working agreement

- One owner per artifact; A integrates reviewed work.
- Browser experiments run serially to avoid cache contamination.
- C owns the evidence definition; D can block misleading claims.
- Any team member can stop release for authorization, privacy or player-safety risk.

9. Delivery plan

From validated mechanism to validated product

Gate	Outcome	Status
1 — Problem and scope	Web-only, browser-native direction locked	Complete
2 — Browser mechanism	Parent request reused by later iframe locally	Complete, title/browser scoped
3 — Safe orchestration	Authorization, governor, exact manifest and fallback	Simulated prototype complete
4 — Staging adapter	Approved configuration and exact title manifest	Next when access arrives
5 — Causal experiment	Isolated CONTROL/TREATMENT with repeat runs	Pending staging
6 — Product decision	Benefit, hit rate and wasted-byte thresholds pass	Pending evidence
7 — Demo and submission	Truthful narrative, reviewer access and freeze audit	Final gate

Success is not “requests were sent”

Success means the predicted title was selected, exact warmed objects were reused, player readiness improved, and speculative cost stayed within policy.

10. Round-one decision

Approve the direction—and validate the outcome next

The proposition

Select one likely next game and prepare a small, exact, authorized set of static assets before the click. Then launch the unchanged provider game normally.

Why proceed

- PSK usage creates repeated opportunities to prepare the next title.
- Historical HARs show the value of warm browser state.
- Local evidence confirms the core parent-to-iframe cache mechanism.
- The design is bounded, reversible and authorization-first.
- Staging access will enable the missing full-title causal proof.

The honest commitment

We are not claiming a production improvement, an 86% prototype gain or a 6-to-1-second result today. We are asking to proceed with a design whose mechanism is locally validated and whose product impact has a clear staging test.

> **Round-one ask:** approve the browser-native prefetch direction, the A/B/C/D execution split and the staging validation gate.

Evidence notes

- `MEASURED` means directly observed or calculated in the named data/capture/experiment.
- `FEG-PROVIDED` means supplied through organiser data or communication.
- `SIMULATED` means a local fixture, policy or UI used for controlled testing.
- `UNKNOWN` means approved evidence does not yet support the claim.

Sources: `CODE.md`, `Context/FINAL-PLAN.md`, `docs/EVIDENCE-STATUS.md`, `docs/HAR-MILESTONE.md`, `docs/LOCAL-CACHE-REUSE.md`, `docs/DECISIONS.md` and `docs/STAGING-SANDBOX.md`.

Raw HARs, provider assets, credentials and player-level data are excluded from this presentation.